CHAITHANYA VEMURI

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Portfolio | GitHub | LinkedIn

PROFILE

I build Python and SQL data pipelines and the infrastructure that delivers ML services. My independent projects include a restartable 411.7M-event pipeline and a controlled GKE lab where I verified API delivery, monitoring and rollback.

TECHNICAL SKILLS

Data engineering: Python, SQL, DuckDB, Parquet, PostgreSQL, dbt, PySpark, Redpanda.

ML systems: PyTorch, scikit-learn, MLflow, FastAPI, offline evaluation.

Platform lab: GCP/GKE, Kubernetes, Terraform, Docker, Helm, Argo CD, GitHub Actions, Prometheus/Grafana.

SELECTED ENGINEERING PROJECTS

FulfillAI | GKE API Delivery & Recovery Lab — GitHub

GCP/GKE | Terraform | Kubernetes | Helm | GitHub Actions | Argo CD

- Provisioned **GKE Autopilot**, VPC and Artifact Registry with Terraform. Used GitHub Actions and Workload Identity Federation for keyless publishing, then deployed pinned image digests via Helm and Argo CD.
- Verified API health and metrics in Prometheus/Grafana and tested Alertmanager-to-Slack routing. Diagnosed an intentional **ImagePullBackOff**, rolled back, and verified Argo CD recovery.
- Built PostgreSQL/dbt data pipelines for **50K synthetic orders** and verified Redpanda/PySpark checkpoint recovery into PostgreSQL.

REES46 V2 | Recommendation & Data Infrastructure — GitHub

Python | SQL | DuckDB | Parquet | FastAPI | PostgreSQL | MLflow

- Processed **411.7M public events** into Parquet. Replaced a global DuckDB aggregation that exhausted temporary storage after a ~30.6 GiB spill with restartable monthly partitions and 64 session hash buckets.
- Improved offline **MRR@20 from 0.02443 to 0.03556** over popularity on **5,000 held-out users**. Trained on 1.50M interactions and exposed the ranking bundle through FastAPI with a popularity fallback.

OTTO | MSc Dissertation in Forecasting — GitHub

Python | PyTorch | Time Series | Chronological Validation

- Streamed **216.7M OTTO click, cart and order events** in one pass into 672 hourly observations. Benchmarked eight 24-hour forecasting methods across three chronological folds and five neural seeds.
- Selected a weekly seasonal baseline at **13.93% mean CV WAPE** versus 17.67% for the PyTorch Transformer. Reached **9.80% cart WAPE and 11.34% order WAPE** on a separate 96-hour holdout.

TRAINING

IntrnForte | Self-directed Python training 02/2024 - 07/2024

EDUCATION

Gisma University of Applied Sciences 08/2025 - 01/2027 (expected)

MSc Data Science, AI & Digital Business Berlin, Germany

Dissertation: Hourly cart and order forecasting from OTTO event logs.

Sri Venkateswara College of Engineering 10/2021 - 05/2025

BTech Electronics & Communication Engineering Tirupati, India

LANGUAGE SKILLS

English: IELTS Academic

German: Currently studying